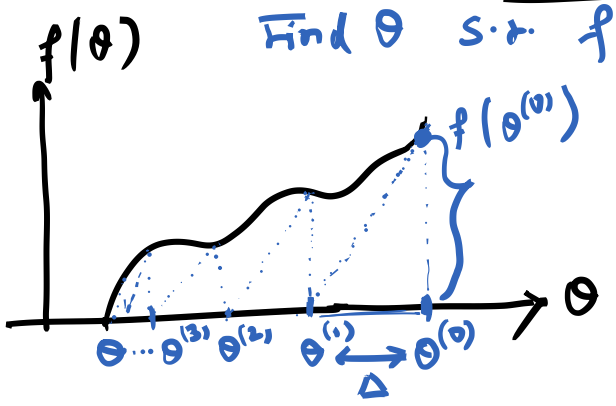
Newton's Method



Find θ s.t. $f(\theta) = 0$

$$\theta^{(1)} = \theta^{(0)} - \Delta$$

$$\text{slope} = f'(\theta^{(0)}) = \frac{f(\theta^{(0)})}{\Delta}$$

$$\Delta = \frac{f(\theta^{(0)})}{f'(\theta^{(0)})}$$

$$\theta^{(1)} = \theta^{(0)} - \frac{f(\theta^{(0)})}{f'(\theta^{(0)})}$$

$$\theta^{(t+1)} := \theta^{(t)} - \frac{f(\theta^{(t)})}{f'(\theta^{(t)})}$$

Let $f(\theta) := \ell'(\theta)$

$$\theta^{(t+1)} := \theta^{(t)} - \frac{\ell'(\theta^{(t)})}{\ell''(\theta^{(t)})}$$

$$\theta^{(t+1)} := \theta^{(t)} - H^{-1} \nabla_\theta \ell(\theta^{(t)}) \qquad \left[H_{ij} = \frac{\partial^2 \ell}{\partial \theta_i \partial \theta_j} \right]$$

Newton's method for Logistic Regression:-

$$\theta^{(t+1)} := \theta^{(t)} - H^{(t)^{-1}} \nabla_\theta (\ell)$$

$$g^{(t)} = \nabla_\theta (\ell) = -\sum_{i=1}^{n} \left(y^{(i)} - h_\theta(x^{(i)}) \right) x^{(i)}$$

$$H = \frac{\partial^2 \ell}{\partial \theta \partial \theta^T} = \frac{\partial}{\partial \theta} \left[\frac{\partial \ell(\theta)}{\partial \theta} \right]^T = \frac{\partial g^T}{\partial \theta}$$

$$= -\frac{\partial}{\partial \theta} \sum_{i=1}^{n} \left(y^{(i)} - h_\theta(x^{(i)}) \right) x^{(i)^T}$$

$$= * \sum_{i=1}^{n} \frac{\partial h_\theta(x^{(i)})}{\partial \theta} x^{(i)^T}$$

$$-\frac{\partial h_\theta(x^{(i)})}{\partial \theta} = \frac{\partial}{\partial \theta}\left[\frac{\exp(\theta^T x^{(i)})}{1+\exp(\theta^T x^{(i)})}\right]$$

$$= h_\theta(x^{(i)})\left[1 - h_\theta(x^{(i)})\right] x^{(i)}$$

$$H = \sum_{i=1}^{n} h_\theta(x^{(i)})\left[1 - h_\theta(x^{(i)})\right] x^{(i)} x^{(i)T}$$

$$H = X^T S X \quad \left[\begin{array}{l} S \text{ is a diagonal matrix.} \\ S_{ii} = h_\theta(x^{(i)})\left[1 - h_\theta(x^{(i)})\right] \end{array}\right.$$

$$\theta^{(t+1)} = \theta^{(t)} - H^{-1} g^{(t)}$$

$$= \theta^{(t)} - \left(\vec{x}^T S^{(t)} \vec{x}\right)^{-1} \vec{x}^T \left(h_\theta(\vec{x})^{(t)} - \vec{y}\right)$$

$$= \theta^{(t)} + \left(x^T S^{(t)} x\right)^{-1} x^T \left(y - h_\theta(x)^{(t)}\right)$$

$$= \left(x^T S x\right)^{-1}\left[\left(x^T S^{(t)} x\right)\theta^{(t)} + x^T\left(y - h_\theta(x)^{(t)}\right)\right]$$

$$= \left(x^T S x^{-1}\right) x^T\left[S^{(t)} x\,\theta^{(t)} + y - h_\theta(x)^{(t)}\right]$$

$$= \left(x^T S x^{-1}\right) x^T S^{(t)}\left[x\,\theta^{(t)} + S^{(t)^{-1}}\left(y - h_\theta(x)^{(t)}\right)\right]$$

$$= \left(x^T S x^{-1}\right) x^T S^{(t)} \hat{y}^{(t)}$$

$$\hat{y}^{(t)} = x\,\theta^{(t)} + S^{(t)^{-1}}\left(y - h_\theta(x)^{(t)}\right)$$

$$\sum_{i=1}^{n} S^{(i)(t)}\left(\hat{y}^{(i)} - \theta^T x^{(i)}\right)^2$$

- Update rule dynamically adjusts the step size by scaling the gradient with inverse Hessian

- Newton's method converges faster for convex problems as compared to GD.

- Computationally expensive.

Multi-class classification:

$y \in \{1, 2, \dots k\}$

$p(y|x;\theta)$ multinomial distribution

$\phi_1, \dots \phi_k$

$\phi_1 = p(y=1|x;\theta_1)$

$\phi_2 = p(y=2|x;\theta_2)$

$\vdots$

$\phi_k = p(y=2|x;\theta_k)$

$\sum_{i=1}^{k} \phi_i = 1$

$\Rightarrow \theta_1^T x \dots \theta_k^T x$

$= \theta_j^T x$ not necessarily between $[0,1]$

- Their summation is not necessarily 1.

$$\text{Softmax}(t_1, \dots t_k) = \begin{bmatrix} \frac{\exp(t_1)}{\sum_{j=1}^{k} \exp(t_j)} \\ \vdots \\ \frac{\exp(t_k)}{\sum_{j=1}^{k} \exp(t_j)} \end{bmatrix}$$

$t_j = \theta_j^T x$